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*The relative influence of morphosyntax and phonology on the
mutual intelligibility of Norwegian and Danish*

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Abstract

Norwegian and Danish are closely related languages and are well-known to have a high degree of mutual intelligibility. The difficulty of comprehension of one of these languages for native speakers of the other is driven by a number of factors: primary amongst these are phonological and morphosyntactic differences. By presenting native Norwegian speakers with audio stimuli constructed variously with Norwegian or Danish phonology and morphosyntax, we perform a statistical analysis of the relative effect of these factors.

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1 Problem and hypothesis

1.1 Mutual Intelligibility and the relative influence of Lexicon, Morphosyntax, and Phonology

For speakers of closely related languages, it is frequently possible to communicate without using the language of the other; this well-studied phenomenon is called ‘mutual intelligibility’, specific cases of which include ‘semicommunication’ (Haugen 1966) and ‘receptive multilingualism’ (Braunmüller and Zeevaert 2001).

A number of factors have been proposed to explain the intelligibility of closely related languages; the most significant of these (Gooskens 2007) are

- the listener’s attitude towards the language,
- the listener’s contact with the language and other languages,
- linguistic distance to the listener’s language.

Of particular interest for this present work is the third factor. Linguistic distance is generally difficult to measure; primary contributions are via lexical, morphosyntactic, and phonological differences (Gooskens and Swarte 2017). Languages having a high degree of similarity on one or more of these can generally be expected to demonstrate some degree of mutual intelligibility, but this has proven very hard to predict.

1.2 Mutual Intelligibility of Nordic Languages

Nordic languages, and particularly the Continental Scandinavian languages Swedish, Danish, and Norwegian, demonstrate a very high degree of mutual intelligibility (Delsing and Lundin-Åkesson 2005; Torp 1998). Interestingly the primary impediments to mutual intelligibility are different among the three languages: Norwegian-Danish is considered to be primarily limited by phonological differences, Norwegian-Swedish by lexical differences, and Danish-Swedish by both lexical and phonological differences.

Previous investigations into lexical similarities and phonological differences between Norwegian and Danish are plentiful. Gooskens (2007) reports 0-1.2% of non-cognate words between Norwegian and Danish written texts. A comparison of phoneme inventories between the languages shows significant differences, for example Norwegian is generally reported to have 9 distinctive long vowels while Danish has 13; simultaneously (eastern dialects of) Norwegian has significantly more consonant phonemes due to the various retroflex sounds that Danish does not have. See Kristoffersen (2000) for the phonetic inventory of ‘Standard Norwegian’ and Basbøll (2005) for that of ‘Standard Danish’.

Notably, many references (see Holmberg and Platzack (2008)) assert that grammatical differences, that is differences in morphosyntax, are negligible between Norwegian and Danish to the degree that they will not impact mutual intelligibility. However, Hilton et al. (2013) carried out a comparative statistical analysis of the effects of phonological and morphosyntactic differences on intelligibility of Norwegian for native Danish speakers, and concluded that while phonological differences were more significant, that morphosyntactic differences had a measurable effect.

1.3 Relative effect of phonology and morphosyntax on the intelligibility of Danish for native Norwegian speakers

This study seeks to measure the effect of morphosyntactic differences between Norwegian and Danish on native Norwegian speakers’ ability to understand Danish. Given the significant lexical

similarities, this will be accomplished as a comparison with the effect of phonological differences. This is a partial replication and direct follow up to Hilton et al. (2013), wherein the relative effect of morphosyntax and phonology on Danish speakers' ability to understand Norwegian was studied.

The primary hypothesis is that morphosyntactic differences will measurably impede both comprehension and response time, but to a lesser degree than phonological differences. Moreover, given that Norwegian exhibits a large spectrum of dialectal differences (that is expressed at the lexical, phonological, and morphosyntactic levels) that native speakers are accustomed to accounting for, it is expected that both phonological and morphosyntactic differences may have even less of an effect on native Norwegian speakers' comprehension of Danish than the measured effects of native Danish speakers' comprehension of Norwegian in Hilton et al. (2013).

2 Methodology

The experiment consists of a series of audio stimuli presented to study participants; participants respond after each stimulus if they consider it to be plausible or not and their response times are measured.

Considering the lexical similarities of Norwegian and Danish, the stimuli were chosen such that all words with semantic content were cognates. This was done to remove consideration of lexical differences entirely from the study.

In order to construct the stimuli, a number of morphosyntactic constructions believed to be potentially problematic for Norwegians in Danish were identified. Notably the differences between the languages are not symmetric; frequently a pair of constructions will both be valid in one language while only one is permissible in the other, but we found that it was much more frequent that Norwegian demonstrated flexibility that Danish did not. In many cases where the corresponding constructions between the languages are different the standard forms, the Danish construction is considered permissible in Norwegian either archaically or dialectically, while the opposite is not true. Because of these asymmetries, none of the constructions used in this study were considered by Hilton et al. (2013).

2.1 Constructions identified for the experiment

We collect the six constructions that were identified for the study here. For each, example test sentences used in the study are included and salient features bolded. Norwegian is the first line of each example and is written always in Bokmål Norwegian; Danish is the second line of each example.

2.1.1 Time constructions using på/i

Both languages use the prepositions *på* (roughly 'on') and *i* (roughly 'in'), which serve a variety of functions. The use of these prepositions in time constructions varies between the two languages. In particular, in phrases measuring time until the next hour (e.g. "quarter to five" meaning 4:45) and in phrases measuring time intervals in the past in negative sentences, Norwegian uses the preposition *på*, while Danish *i*.

(1) *'Time preposition' Construction*

a. *'Time interval in negated sentence' case*

Jeg har ikke sett ham **på** to år.

Jeg har ikke set ham **i** to år.

'I haven't seen him **for** two years.'

- b. ‘Time to next hour’ case
 Selskapet begynner kvart **på** sju.
 Selskapet begynder kvart **i** syv.
 ‘The party begins quarter **to** seven.’

2.1.2 Subordination via som/der

In both languages, relative clauses are introduced with the subjunction *som* (‘that’/‘which’/‘who’), however in Danish if the noun being modified is the grammatical subject of the sentence then *der* (‘that’) is preferred. The use of *der* in this context is considered ungrammatical or archaic in Norwegian.

(2) ‘Subject relative clause’ Construction

En fisk **som** er velutdannet leser avisen.
 En fisk **der** er veluddanet læser avisen.
 ‘A fish **who** is well educated reads the newspaper.’

2.1.3 Perfect auxilliary verb ha/er

Both languages have the verbs *å være* / *at være* (‘to be’) and *å ha* / *at have* (‘to have’), which serve various semantic and syntactic functions that are largely equivalent in the two languages. In the construction of the compound perfect verb forms Norwegian invariantly uses *å ha* as an auxiliary while Danish variously uses either *at have* or *at være* depending on the verb - there are even some verbs for which both are valid with semantic differences.

(3) ‘Perfect auxilliary’ Construction

Harald **har** **ankommet** lufthavnen.
 Harald **er** **ankommet** lufthavnen.
 ‘Harald **has** **arrived** at the airport.’

2.1.4 Plural formation

Norwegian plural formation is highly regular via suffixes.

	Singular	Plural
Indefinite	et/en/ei (<i>root</i>)	(<i>root</i>)-er
Definite	(<i>root</i>)-et/-en/-a	(<i>root</i>)-ene

Table 1: Norwegian plural formation

Danish, however, exhibits two regular patterns, and the applicable pattern for a given word cannot easily be predicted.

	Singular	Plural		Singular	Plural
Indefinite	et/en (<i>root</i>)	(<i>root</i>)-er	Indefinite	et/en (<i>root</i>)	(<i>root</i>)-e
Definite	(<i>root</i>)-et/-en	(<i>root</i>)-erne	Definite	(<i>root</i>)-et/-en	(<i>root</i>)-ene

Table 2: Danish plural formation

For the purposes of the experiment, we considered two constructions: a Definite Plural construction when Danish chooses the *-erne* suffix:

(4) *'Definite Plural' Construction*

Min elev mistet **bøkene**.
Min elev mistede **bøgerne**.

'My student lost **the books**.'

and an Indefinite Plural construction where the difference occurs when Danish chooses the *-e* suffix:

(5) *'Indefinite Plural' Construction*

Jorden har tre **soler**.
Jorden har tre **sole**.

'The Earth has three **suns**.'

2.1.5 Definiteness marking of noun phrases

Both Norwegian and Danish mark definite nouns via suffix, but noun phrases including an adjective or a demonstrative pronoun are handled differently; Norwegian performs double marking, including a definite article or demonstrative pronoun followed by the definite form of the adjective and finally the head noun marked as definite via suffix. Danish only performs single marking, retaining the definite article while the head noun takes its root form.

(6) *'Definite noun phrase' Construction*

a. *'Adjective' case*

Hun bor i **det grønne huset**.
Hun bor i **det grønne hus**.
'She lives in **the green house**.'

b. *'Demonstrative pronoun' case*

Den mannen er min mor.
Den mand er min mor.
'**That man** is my mother.'

2.2 Design of Stimuli

For each of the 6 constructions discussed in section 2.1, 6 sentences were constructed - 3 intended to be plausible, and 3 implausible - for a total 36 test sentences. They were chosen such that all words with semantic content were cognates in Norwegian and Danish as previously discussed, and such that there were minimal differences other than the specific construction being tested. All sentences were translated into standard Danish and Bokmål Norwegian.

In order to distinguish between the effects of phonological and morphosyntactic differences on intelligibility, each test sentence was recorded variously using Danish and Norwegian morphosyntax and using Danish and Norwegian phonology, for a total of 4 recorded stimuli per test sentence and an overall total 144 recorded stimuli. That is, each test sentence was recorded in the following 4 configurations:

- NpNm - Norwegian phonology, Norwegian morphosyntax
- NpDm - Norwegian phonology, Danish morphosyntax

- DpNm - Danish phonology, Norwegian morphosyntax
- DpDm - Danish phonology, Danish morphosyntax

The complete list of test sentences and their translations can be found in appendix [A](#).

2.3 Practical considerations for recording stimuli

The stimuli were recorded by 2 native speakers each of Norwegian (speakers ‘e’ and ‘p’) and Danish (speakers ‘m’ and ‘s’). Each speaker created 36 recordings; 18 each in the configurations Nm and Dm, where the phonology was that of their native language. The recordings made by each speaker were distributed evenly across all stimuli so as to minimize the effect of the individuals’ recorded voices on the experiment.

When recording the morphosyntax of their native language, the speakers were given no special instructions. The “mixed cases” NpDm and DpNm were somewhat more difficult, and the speakers were asked to speak as naturally as possible, but making sure to express the salient morphosyntactic feature as it is expressed in the morphosyntax they were recording (that is, not their native). Several native speakers of each language were asked to listen to the recordings and agreed that the recordings all sounded native, but no detailed analysis was undertaken of the stimuli.

Recordings were volume normalized and cropped to cut any empty time from the beginning and end, but were otherwise unmodified.

2.4 Execution

The experiment was conducted using the Gorilla experiment platform (gorilla.sc), wherein both the experiment was designed and the data collected.

Participants wore headphones and sat in a relatively quiet, but otherwise uncontrolled, area for the experiment. All participants used the same laptop (a Windows Surface Pro laptop running Linux Mint) and wore the same headphones (a Soundcore Q20i wireless overear headphone with external noise-cancelling function). Testing via the online Latency Tester <https://nullvoxbopuli.github.io/latency-tester/> measured an average 214ms latency in the headphone audio over 4 tests; this can affect the absolute response times measured, but not the effect size.

Participants were first presented with an instruction screen, given an example of a plausible and implausible sentence, and then practised using 4 sample plausibility determination tasks for which the stimuli were recorded in configuration NpNm. After confirming they understood the instructions, participants were then presented with 36 plausibility determination tasks - one for each of the test sentences.

For each plausibility determination task participants were presented with an audio stimulus. For the duration of each, a progress bar indicated when the stimulus would end. Then the screen displayed the choice between plausible (marked green) and implausible (marked red); participants responded by pressing the corresponding color marked keys (‘f’ and ‘j’ respectively on a standard Norwegian keyboard). Participants were able to then start the next task by pressing the spacebar, but could neither stop nor repeat the stimulus once playback had begun.

The order of the test sentences was completely randomized for each participant. For each test sentence, which of the 4 stimuli configurations they were presented with for each sentence was also randomized; this randomization was such that a uniform distribution across the configurations - 9 of each - was presented to each participant.

Data collection via Gorilla included detailed data on response correctness, response times, the order in which the test sentences were presented, and which stimulus was presented. Response time was measured from the end of the audio stimulus until the response of either ‘plausible’ or ‘implausible’ via the keyboard.

3 Experimental Results

All statistical analysis was carried out using the R version 4.6.0 statistical software in RStudio. The full code is available in appendix B.

The participants were all native Norwegian speakers that had never lived in Denmark and had no formal education in Danish. All (N=22) lived in the Bergen area at the time of the experiment, but originated variously from Vestland (N=13), Akershus (N=2), Oslo (N=2), Østfold (N=2), Buskerud (N=1), Innlandet (N=1), and Trøndelag (N=1). Participants included 12 men, 9 women, and 1 non-binary person. Ages ranged from 21 to 63, with mean 33.4. All participants had completed high school, with (N=3) holding a Bachelor’s degree, (N=6) holding a Master’s degree, and (N=2) holding a doctorate. All were recruited directly by the primary investigator.

Because of the small sample size, the demographic information was not analyzed.

3.1 Dataset trimming for difficult sentences

It was not anticipated that any participants would correctly answer all 36 stimuli. Correct response rates ranged from 61.1% (with 22 correct responses) to 97.2% (with 35 correct responses) with mean 81.3%.

We observed that several of the test sentences appear difficult for participants to interpret. A reasonable test sentence should be correctly interpreted by most participants if presented in Norwegian, and so any sentences that in configuration NpNm were incorrectly answered more than once or which were only presented once and answered incorrectly were classified as difficult, and all primary results are presented with all difficult sentences (in all configurations) removed. These were:

Index	Sentence	Plausibility	n	Error
20	We get home from school at 7.40 in the morning.	NP	1	1
11	The schools will not teach adults.	P	5	4
3	Ten to eighteen (17.50) we go to bed.	P	8	4
15	Our many dogs will not bark at strangers.	P	7	3
34	That man is my mother.	NP	8	2

Table 3: Difficult sentences

Here ‘Index’ is the sentence index used internally, ‘Plausibility’ is the intended interpretation (P plausible, NP not plausible), ‘n’ is the number of participants who were presented with the sentence in configuration NpNm, and ‘Error’ is the number of participants who answered incorrectly in configuration NpNm.

After removal of the 5 difficult sentences, there were 682 observations (22 participants, 31 stimuli). The correctness distribution for the non-difficult sentences ranged from 61.3% (with 19 correct) to 96.8% (with 30 correct) with mean 83.9%.

3.2 Correctness Analysis

The percentage of correct replies across configurations and participants for the non-difficult dataset was analyzed via two-way ANOVA. ANOVA (ANalysis Of VAriance) tests specifically determine the effect of independent categorical variables on a continuous independent variable (Montgomery 2020). Here the phonology P and morphosyntax M are treated as categorical variables and the percentage correct responses for a given participant is the independent variable.

It is established in the literature that raw percentage data is unsuitable for statistical analysis, as the size of the variance is related to the size of the mean; per Studebaker (1985) a ‘rationalized arcsine transform’ is appropriate to prepare such data for analysis. We chose to use the second transformation discussed in that paper, with consideration for our sample size.

For each participant and each configuration we determined the correctness count x = ‘number of correct responses’ and total count n = ‘number of responses’. These were then transformed as

$$T = \arcsin \sqrt{\frac{x}{n+1}} + \arcsin \sqrt{\frac{x+1}{n+1}}$$

$$R = 46.47324337 \cdot T - 23$$

The resulting data was analyzed via two-way ANOVA using the R software packages ‘stats’ and ‘car’. Both phonology ($F(1) = 18.717$) and morphosyntax ($F(1) = 16.762$) were reported to have a significant effect ($p < 0.001$). We did also analyze the interaction and find none, ($F(1) = 0.209$, $p = 0.65$).

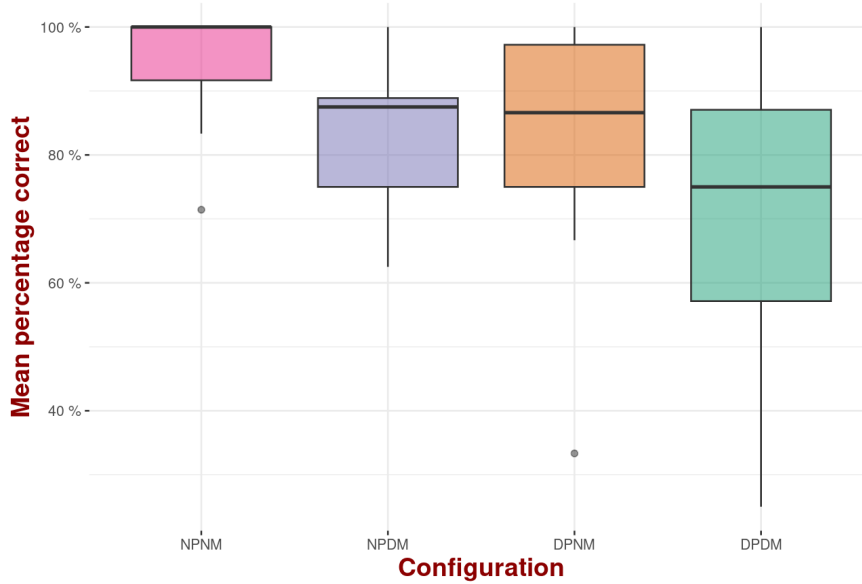


Figure 1: Mean percentage correct responses by configuration

3.3 Controlling for correctness in Response Time analysis

Response time is a richer observation than correctness, and so several analyses were undertaken. We first observe via a one-way ANOVA test over correct/incorrect responses that response time for incorrect responses was greater ($F(1) = 21.63$, $p < 0.001$). This is observed visually in fig. 2.

In order to control for the response time difference between correct and incorrect responses the non-difficult dataset with only correct responses is used for the remainder of the response time analysis. This dataset consists of 572 observations.

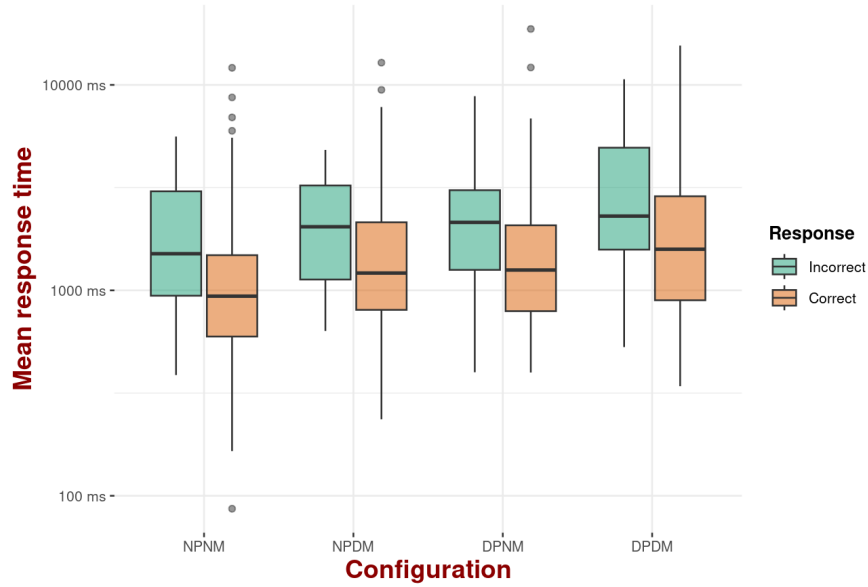


Figure 2: Mean response time by configuration, comparison for Correct/Incorrect responses

3.4 Effect of controlled variables

We use one-way ANOVA testing to study the effect of the speakers and individual morphosyntactic constructions on both correctness and response time.

3.4.1 Speaker effect analysis

Here we confirm that differences in the speakers' voices did not result in an effect on correctness or response time.

Performing two one-way ANOVA analyses of correctness over the speakers, one for the data with only the Norwegian speakers ('e' and 'p') and one for the data with only the Danish speakers ('m' and 's') we obtain that neither had a significant effect: $F(1) = 0.374$, $p = 0.544$ for the the Norwegian speakers, $F(1) = 0.026$, $p = 0.874$ for the Danish speakers.

Analysis of response time also concludes that neither had a significant effect: $F(1) = 0.824$, $p = 0.365$ for the the Norwegian speakers (fig. 3), $F(1) = 2.311$, $p = 0.13$ for the Danish speakers (fig. 4).

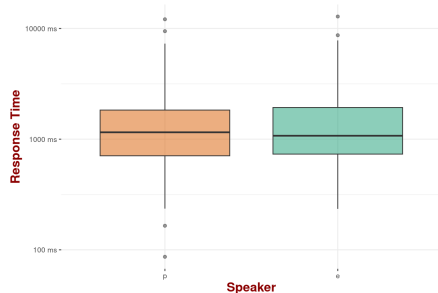


Figure 3: Effect of Norwegian speakers on response time

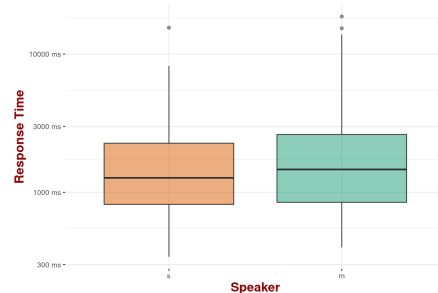


Figure 4: Effect of Danish speakers on response time

3.4.2 Construction effect analysis

We also run one-way ANOVA tests for correctness and response time over the six morphosyntactic constructions discussed in section 2.1. This confirms significant effect with $F(5) = 3.87$, $p < 0.01$ for correctness and $F(5) = 2.3465$, $p < 0.05$ for response time (fig. 5). The mean correctness and response times for each construction are collected in fig. 6.

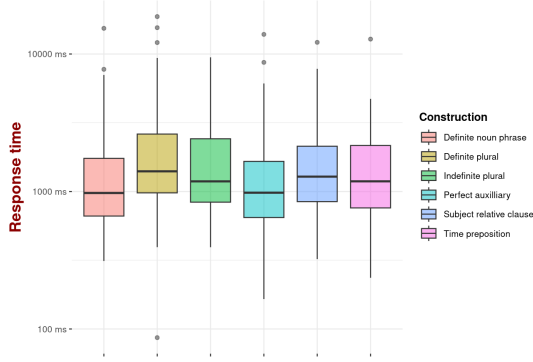


Figure 5: Response time versus construction

Construction	Corr.	R. time (ms)
Definite plural	0.782	2456.8
Perfect auxilliary	0.795	1535.6
Time preposition	0.818	1679.0
Subject relative clause	0.833	1903.2
Indefinite plural	0.864	1740.1
Definite noun phrase	0.945	1665.3

Figure 6: Mean correctness and response time by construction

3.5 Linear mixed-effect regression

We now build a regression model for response time R . The basic linear regression model is

$$R = m_P P + m_M M + a + \epsilon \quad (1)$$

where P (phonology) and M (morphosyntax) are logical variables taking the values 0 for Norwegian and 1 for Danish. We use linear mixed-effect regression (Finch et al. 2019), whereby the intercept a is estimated uniquely for each participant (id), each sentence (index), and each morphosyntactic construction (construction); the variable names as they appear in the data are indicated in parenthesis. In particular we used a nested analysis for the sentences and constructions (index : construction).

The $\epsilon \sim N(0, \sigma)$ term is a normally distributed error, accounting for noise and any effects not being modelled. The theoretical values of the slopes m_P and m_M describe the effect (in milliseconds) of introducing Danish phonology and morphosyntax, respectively.

Linear mixed-effect modelling via R software package ‘lme4’ function ‘lmer’ estimates $m_P \approx 579.09$ ms, $m_M \approx 635.08$ ms, both having effect with significance $p < 0.001$ (fig. 7).

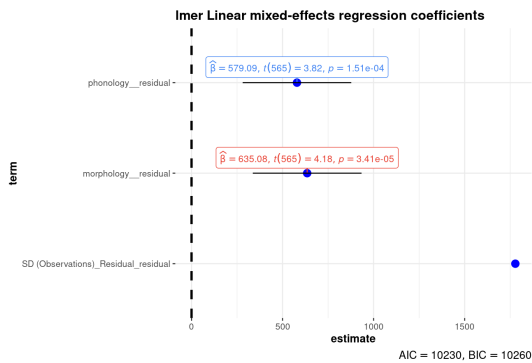


Figure 7: Estimates for m_P and m_M with 95% confidence intervals.

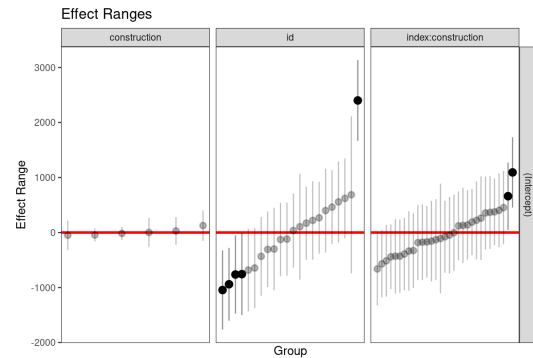


Figure 8: Comparison of random effects

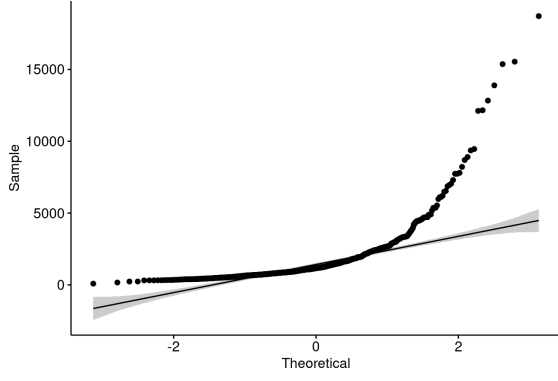


Figure 9: Q-Q plot for linear data

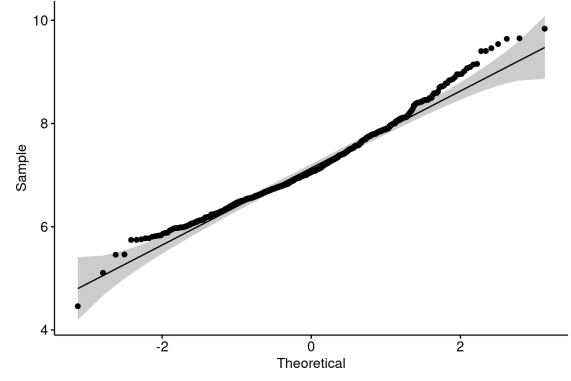


Figure 10: Q-Q plot for log-data

However, a Q-Q plot (fig. 9) indicates that outliers may be too far from normalcy to be appropriate for such analysis, which is supported by a Shapiro-Wilk test ($p < 0.001$ for normalcy). Moreover, a simulation of the intercepts by participants (fig. 8) also indicates that there are outliers in the data.

3.5.1 Accounting for non-normal data

To account for the potential non-normalcy of the data, we have two approaches:

Robust linear mixed-effect regression

By using the R software package ‘robustlmm’ function ‘rlmer’ we can perform a robust fitting of the linear mixed-effect model; the essential idea is that the dataset is weighted against the effects of ‘contamination’, see Koller (2016) for details of the implementation. With this model we have estimates $m_P \approx 379.7959$ ms, $m_M \approx 370.2453$ ms, both having effect with significance $p < 0.001$.

Log transformation

By transforming the response data using a logarithm the effect of outliers is minimized. This is a standard practice for response time data analysis, see Whelan (2008) for a discussion. We also observe that the Q-Q plot (fig. 10) for the log-data is significantly more promising.

The model here becomes

$$\ln(R) = m_P P + m_M M + a + \epsilon \quad (2)$$

for which we obtain estimates $m_P \approx 0.3097481$, $m_M \approx 0.3057751$, both having effect with significance $p < 0.001$. We note that these parameters can no longer be directly interpreted as a millisecond increase in response time; the usual interpretation is as a percent increase, however since the variables are logical it is possible to compute millisecond response time increase, which we do in section 3.6.

It is possible to apply the robust model to the log-data. The estimated coefficients are very similar, with $m_P \approx 0.3037952$, $m_M \approx 0.2830915$, both having effect with significance $p < 0.001$.

3.6 Estimated coefficient interpretation and Equivalence analysis

We collect in table 4 a summary of the model results.

	Linear		Log-Linear	
	lmer	rlmer	lmer	rlmer
(Intercept)	1320.512*** (241.331)	1154.776*** (144.841)	6.910*** (0.107)	6.909*** (0.109)
phonology	579.091*** (151.787)	379.796*** (72.413)	0.310*** (0.052)	0.304*** (0.051)
morphosyntax	635.082*** (152.027)	370.245*** (72.565)	0.306*** (0.052)	0.283*** (0.051)
SD (Intercept index : construction)	521.129	311.870	0.259	0.257
SD (Intercept id)	816.233	468.369	0.380	0.341
SD (Intercept construction)	143.335	142.765	0.075	0.113
SD (Observations)	1778.251	824.872	0.603	0.578
Num.Obs.	572	572	572	572
R^2 Marg.	0.042	0.063	0.074	0.074
R^2 Cond.	0.265	0.374	0.420	0.415

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 4: Summary of regression estimates

The estimated coefficients of the linear models can be interpreted as a direct estimate of the millisecond increase of response time, while the estimated coefficients of the log-linear models can be approximately interpreted as indicating a $(e^m - 1) \cdot e^a$ millisecond increase of response time; we summarize these and the relative effect (their ratios) in table 5.

	Linear		Log-Linear	
	lmer	rlmer	lmer	rlmer
phonology	579.091	379.796	364.242	355.712
morphosyntax	635.082	370.245	358.787	327.513
relative effect	0.912	1.03	1.01	1.07

Table 5: Estimated millisecond increase in response time by model and relative effect

We present also a visual comparison of the confidence intervals (fig. 11).

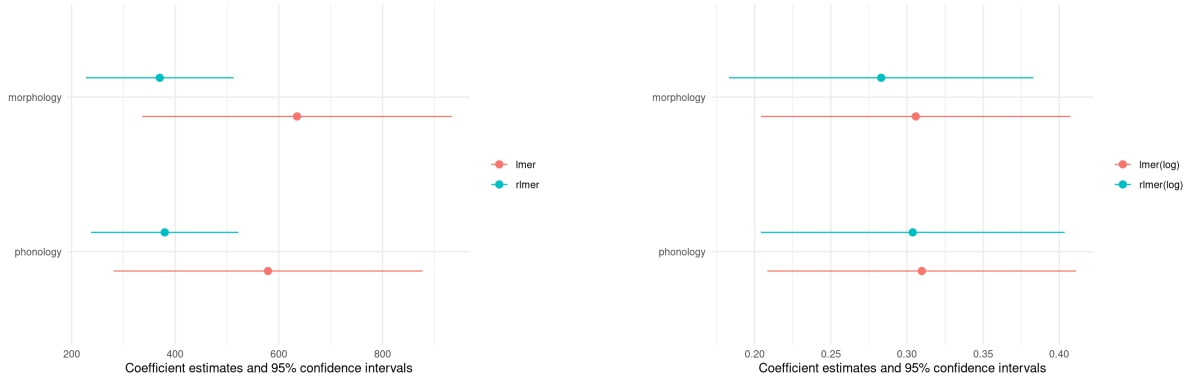


Figure 11: Coefficient estimates and confidence intervals

Equivalence testing

We observe that the relative effect of morphosyntax and phonology is roughly 1 in all of our models, prompting us to undertake an equivalence analysis. To do this, we build a new linear regression model in the form

$$R = b_1(P + M) + b_2(P - M) + a + \epsilon.$$

Comparing with eq. (1), it follows that $b_2 = \frac{1}{2}(m_P - m_M)$, and so we perform a t-test on b_2 ; if the null hypothesis $H_0: b_2 = 0$ holds, we must also conclude that $m_P = m_M$. For the log lmer model, we find $t = 0.055$, $p = 0.956$ and conclude that the effects of morphosyntax and phonology are equivalent. Carrying out the analysis under different models gives the same conclusion.

There are other methods of equivalence testing: we also perform an ANOVA comparison of the log lmer model eq. (2) to a restricted log lmer model of the form

$$\ln(R) = m(P + M) + a + \epsilon$$

and conclude the same, with $\chi^2(1) = 0.003$, $p = 0.9565$.

See Robinson and Froese (2004) for a motivating analysis of equivalence tests and Ben-Shachar (2021) for a friendly discussion of their implementation in R.

4 Discussion and Conclusions

4.1 Effect significance

Both the ANOVA analysis of correctness rates (section 3.2) and all of the regression models for response times (section 3.5) consistently determined that both morphosyntax and phonology have statistically significant impact $p < 0.001$. We conclude that both factors are relevant for Norwegian speakers' comprehension and processing of Danish, which agrees with our hypothesis.

We note that we only tested a limited number of morphosyntactic features, and those were specifically chosen as potentially difficult for Norwegian speakers. For samples chosen without consideration for potential difficulty, it is unclear how significant morphosyntactic factors will be while it seems likely that phonological differences will have a constant effect. Further research into the extent of such difficult constructions and the relative frequency of their use in Danish would be necessary to answer this question satisfactorily.

4.2 Construction comparison

Comparison of correctness and response times across the individual morphosyntactic constructions section 3.4 reveals that their effect was unequal; this is not unexpected, but it is of interest to see which have greatest and least effect. With lowest mean correctness (78.2%) and greatest mean response time (2457ms), the ‘Definite plural’ construction appears to have been the most difficult. On the opposite end, the ‘Definite noun phrase’ construction had the highest mean correctness (94.5%) and second least mean response time (1665ms), and so it seems to have been relatively simple. The ‘Perfect auxiliary’ construction also stands out, with the second lowest mean correctness (79.5%) but also lowest mean response time (1536ms), which indicates that it was perhaps harder to interpret than expected.

Collecting data on participants’ confidence in their answers would have provided an interesting measure against which to interpret the apparent lack of correlation between mean response times and mean correctness rates across constructions.

It would perhaps be of interest to further catalog various morphosyntactic constructions that differ between the two languages and attempt to more quantitatively describe the relative effect of these constructions. Controlling for phonological and lexical differences could give insight into how large a gap these constructions imply between the languages.

4.3 Effect strength

To measure the strength of these effects, it is most natural to consider the regression analysis of response times. Note, the R^2 values for all of the regression models were quite low, the highest being the robust log-linear regression with $R^2 = 0.415$. This is reasonable, indicating that the models are not predictive and that there are a number of factors that we have not accounted for. However, this is not a problem for our analysis, as we are interested in measuring the (relative) strength of the effects of changing the morphosyntax and phonology, which can reasonably be expected to be captured by the models.

In all regression models the effect of morphosyntax and phonology were equivalent, as discussed in section 3.6 and can be seen in table 5 and fig. 11. From this we conclude that morphosyntax (in these ‘difficult’ constructions) has a comparable effect to that of phonology on the mutual intelligibility of Danish on Norwegian speakers. This is stronger than our original hypothesis, and is in contrast to previous theory.

In particular, Hilton et al. (2013) found that the effect of morphosyntax on Danish speakers’ comprehension of Norwegian was significant, but weaker than that of phonology. Two possible explanations are that 1) morphosyntax is a greater impediment to mutual intelligibility for Norwegian speakers than it is for Danish speakers, or instead that 2) phonology has a smaller effect on Norwegian speakers than Danish speakers. It is not simple to directly compare the results of these two studies; a future work collecting and uniformly comparing data in both directions could answer this question, but existing literature already demonstrates asymmetry in the mutual intelligibility of the languages and so the result is perhaps not unexpected.

Also, given the breadth of dialects in Norwegian and their varying linguistic distances from Danish, we hypothesize that the effects of both morphosyntax and phonology on Norwegian speakers’ comprehension of Danish may be dependent on their dialect. Because of the small sample size and the significant bias in participants from one region (Vestland), we suggest that further research can be carried out better controlling for this factor.

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A Test Sentences

The test sentences used for creation of the stimuli are collected here. The first line is the English translation, the second the (Bokmål) Norwegian, and the third Danish. The small numbers correspond to the sentence indices used in the data. Sentences in red formed the difficult dataset (section 3.1).

Practice

(1) Plausible

- a. *'The house is built on top of the hill.'*
Huset bygges på toppen av bakken.
- b. *'Kari does not like green tea.'*
Kari liker ikke grønn te.

(2) Implausible

- a. *'The book is made of water.'*
Boken er laget av vann.
- b. *'Two thousand men shared one fish for dinner.'*
To tusen menn delte én fisk til middag.

Plausible

(1) 'Time preposition' Construction

- 1. *'I haven't seen him for two years.'*
Jeg har ikke sett ham på to år.
Jeg har ikke set ham i to år.
- 2. *'The party begins quarter to seven.'*
Selskapet begynner kvart på sju.
Selskapet begynner kvart i syv.
- 3. *'Ten to six we go to bed.'*
Ti på seks går vi i seng.
Ti i seks går vi i seng.

(2) 'Subject relative clause' Construction

- 4. *'The man who ate the food gave me a dour look.'*
Mannen som spiste maten ga meg et surt blikk.
Manden der spiste maden gav mig et surt blik.
- 5. *'He who is most hungry runs the fastest.'*
Den som er mest sulten løper hurtigst.
Den, der er mest sulten, løber hurtigst.
- 6. *'The tree which is tallest is cut first.'*
Treet som er høyest hugges først.
Træet der er højest hugges først.

(3) **‘Perfect auxiliary’ Construction**

7. *‘Harald has arrived at the airport.’*
Harald har ankommet lufthavnen.
Harald er ankommet lufthavnen.
8. *‘Ellie has stayed in the country after her vacation.’*
Ellie har blitt i landet etter sin ferie.
Ellie er blevet i landet etter sin ferie.
9. *‘During the earthquake my shelf has fallen.’*
Under jordskjelvet har min hylle falt.
Under jordskælvet er min hylde faldt.

(4) **‘Definite plural’ Construction**

10. *‘He cleans the streets every night.’*
Han rengjør gatene hver natt.
Han rengør gaderne hver nat.
11. *‘The schools will not teach adults.’*
Skolene vil ikke undervise voksne.
Skolerne vil ikke undervise voksne.
12. *‘My student lost the books.’*
Min elev mistet bøkene.
Min elev mistede bøgerne.

(5) **‘Indefinite plural’ Construction**

13. *‘There are many boats in the seaport.’*
Det er mange båter i havnen.
Der er mange både i havnen.
14. *‘We hang our hats in the entranceway.’*
Vi henger våre hatter i inngangen.
Vi henger vores hatte i indgangen.
15. *‘Our many dogs will not bark at strangers.’*
Våre mange hunder vil ikke gjø på fremmede.
Vores mange hunde vil ikke gø ad fremmede.

(6) **‘Definite noun phrase’ Construction**

16. *‘Even the smallest houses in Norway have a toilet.’*
Selv de minste husene i Norge har et toalett.
Selv de mindste huse i Norge har et toilet.
17. *‘She lives in the green house.’*
Hun bor i det grønne huset.
Hun bor i det grønne hus.
18. *‘People claim that this car is the favorite to win the race.’*
Folk hevder at denne bilen er favoritten for å vinne billøpet.
Folk hævder at denne bil er favorit til at vinde billøbet.

Implausible

(1) **‘Time preposition’ Construction**

19. *‘It hasn’t rained in Bergen for ten years.’*

Det har ikke regnet i Bergen på ti år.

Det har ikke regnet i Bergen i ti år.

20. *'We get home from school at 7.40 in the morning.'*

Vi kommer hjem fra skolen tjue på åtte om morgenen.

Vi kommer hjem fra skole tyve i otte om morgenen.

21. *'The sun rises quarter to one.'*

Solen står opp kvart på ett.

Solen står op kvart i et.

(2) **'Subject relative clause' Construction**

22. *'The box which is largest fits the least.'*

Kassen som er størst rommer minst.

Kassen der er størst rummer mindst.

23. *'A fish who is well educated reads the newspaper.'*

En fisk som er velutdannet leser avisen.

En fisk der er veluddanet læser avisen.

24. *'A book which is written in Arabic is easy to read for everyone.'*

En bok som er skrevet på arabisk er lett å lese for alle.

En bog der er skrevet på arabisk er let at læse for alle.

(3) **'Perfect auxilliary' Construction**

25. *'The refrigerator has gone back to get the lunch bag.'*

Kjøleskapet har gått tilbake for å hente matposen.

Køleskabet er gået tilbage for at hente madposen.

26. *'My cat has gone to the door to open it for the guest.'*

Katten har gått til døren for å åpne den for gjesten.

Katten er gået til døren for at åbne den for gæsten.

27. *'John-Timmy has travelled to the moon by boat.'*

John-Timmy har reist til månen med båt.

John-Timmy er rejst til månen med båd.

(4) **'Definite plural' Construction**

28. *'The museums in Norway do not hang paintings.'*

Museene i Norge henger ikke malerier opp.

Museerne i Norge hænger ikke malerier op.

29. *'In Oslo the buildings are the smallest of all Norwegian cities.'*

I Oslo er bygningene de minste av alle norske byer.

I Oslo er bygningerne de mindste af alle norske byer.

30. *'The boats are sailing over the bridges.'*

Fartøyene seiler over broene.

Fartøjerne sejler over broerne.

(5) **'Indefinite plural' Construction**

31. *'Large flocks of cats are hunting men.'*

Store flokker av katter jakter menn.

Store flokke af katte jager mænd.

32. *'The Earth has three suns.'*

Jorden har tre soler.

Jorden har tre sole.

33. *'There are never boats in the busy seaport.'*

Det er aldri båter i den travle havnen.

Der er aldrig både i den travle havneby.

(6) **'Definite noun phrase' Construction**

34. *'That man is my mother.'*

Den mannen er min mor.

Den mand er min mor.

35. *'That hard stone is as soft as butter.'*

Den harde steinen er formbar som smør.

Den hårde sten er formbar som smør.

36. *'Ellie lives together with that grey cat in a desk.'*

Ellie bor sammen med den grå katten i et skrivebord.

Ellie bor sammen med den grå kat i et skrivebord.

B R Code Reference

This code was compiled on Linux Mint in RStudio running R ver. 4.6.0. The .R file and all data .csv files are available on <https://github.com/mercruccio13/thesis-2026>.

analysis.R

```
library(tidyverse)

cleandata <- function(datafile) {

  rdata <- read.csv(datafile)
  rdata |>
  filter(Response.Type == "response", Display == "Trial") |>
  mutate(
    pid = Participant.Private.ID,
    conf = Spreadsheet..setting,
    stimulusfile = case_when(
      conf == "DPDM" ~ Spreadsheet..DPDM,
      conf == "DPNM" ~ Spreadsheet..DPNM,
      conf == "NPDM" ~ Spreadsheet..NPDM,
      conf == "NPNM" ~ Spreadsheet..NPNM
    ),
    p = str_extract(Spreadsheet..setting, ".(?:=P)"),
    phonology = case_when(
      p == "N" ~ 0,
      p == "D" ~ 1,
    ),
    m = str_extract(Spreadsheet..setting, ".(?:=M)"),
    morphology = case_when(
      m == "N" ~ 0,
      m == "D" ~ 1,
    ),
    index = as.integer(str_extract(stimulusfile, "\\d+")),
    sconst = (((index - 1) %% 18) %/% 3) + 1,
    construction = case_when(
      sconst == 1 ~ "Time preposition",
      sconst == 2 ~ "Subject relative clause",
      sconst == 3 ~ "Perfect auxilliary",
      sconst == 4 ~ "Definite plural",
      sconst == 5 ~ "Indefinite plural",
      sconst == 6 ~ "Definite noun phrase",
    ),
    speaker = str_extract(stimulusfile, ".(?:=\\.mp3$)"),
    correct = as.logical(Correct)) |>
  group_by(pid) |>
  mutate(id = cur_group_id()) |>
  ungroup() |>
  select(id,
    index,
    construction,
    conf,
    phonology,
    morphology,
    speaker,
    Reaction.Time,
    correct)
}
```

```

remove_errdata <- function(datain) {
  errdata <- datain |>
    group_by(index, conf) |>
    summarise(n = n(), err = (n() - sum(correct)), .groups = "drop_last") |>
    filter(conf == "NPNM" & (err > 1 | (err == 1 & n == 1)))
  anti_join(datain,
    errdata,
    by = join_by(index))
}

#Load data
data <- cleandata("data-raw.csv")

#Plot Reaction Time by participant, consider outliers
data |>
  ggplot(aes(group = id, y = Reaction.Time)) +
  geom_boxplot()

#Look for high error rates in NPNM
data |>
  group_by(index, conf) |>
  summarise(pcorr = sum(correct)/n(),
    err = (n() - sum(correct)), n = n(),
    meanrtime = mean(Reaction.Time)) |>
  filter(conf == "NPNM", err > 0) |>
  arrange(pcorr, desc(err))

#Remove sentences with high NPNM error rates
nerrdata <- remove_errdata(data)

#Correctness Analysis
library(car)
library(ggstatsplot)

#Two-way ANOVA for phonology/morphology
mpcorrdata <- nerrdata |>
  group_by(id, conf) |>
  summarise(x = sum(correct),
    xp = x/n(),
    t = asin(sqrt(x/(n()+1))) + asin(sqrt((x+1)/(n()+1))),
    r = 46.47324337*t - 23,
    .groups = "drop_last") |>
  mutate(p = str_extract(conf, "(?=P)"),
    phonology = case_when(
      p == "N" ~ 0,
      p == "D" ~ 1,
    ),
    m = str_extract(conf, "(?=M)"),
    morphology = case_when(
      m == "N" ~ 0,
      m == "D" ~ 1,
    )
  )

summary(aov(r ~ phonology * morphology, data = mpcorrdata))
Anova(aov(r ~ phonology + morphology, data = mpcorrdata), type = 2)

mpcorrdata |>
  ggplot(aes(x = conf, y = xp, fill = conf)) +

```

```

geom_boxplot(alpha = 0.5) +
labs(x = "Configuration", y = "Mean percentage correct") +
scale_y_continuous(label = function(x) {return(paste(x*100, "%"))}) +
scale_x_discrete(limits=rev) +
theme_ggstatsplot() +
theme(legend.position = "none",
      axis.title = element_text(color = "darkred", size = 14)) +
scale_fill_brewer(palette="Dark2")

# One-way ANOVA for Norweigan speakers
mp_speaker_n <- nerrdata |>
  group_by(id, speaker) |>
  summarise(n = n(),
            x = sum(correct),
            xp = x/n(),
            t = asin(sqrt(x/(n()+1))) + asin(sqrt((x+1)/(n()+1))),
            r = 46.47324337*t - 23,
            .groups = "drop_last") |>
  filter(speaker %in% c("e", "p"))
summary(aov(r ~ speaker, data = mp_speaker_n))

mp_speaker_n |>
  ggplot(aes(x = speaker, y = xp, fill = speaker)) +
  geom_boxplot(alpha = 0.5) +
  labs(x = "Configuration", y = "Mean percentage correct") +
  scale_y_continuous(label = function(x) {return(paste(x*100, "%"))}) +
  scale_x_discrete(limits=rev) +
  theme_ggstatsplot() +
  theme(legend.position = "none",
        axis.title = element_text(color = "darkred", size = 14)) +
  scale_fill_brewer(palette="Dark2")

# One-way ANOVA for Danish speakers
mp_speaker_d <- nerrdata |>
  group_by(id, speaker) |>
  summarise(n = n(),
            x = sum(correct),
            xp = x/n(),
            t = asin(sqrt(x/(n()+1))) + asin(sqrt((x+1)/(n()+1))),
            r = 46.47324337*t - 23,
            .groups = "drop_last") |>
  filter(speaker %in% c("m", "s"))
summary(aov(r ~ speaker, data = mp_speaker_d))

mp_speaker_d |>
  ggplot(aes(x = speaker, y = xp, fill = speaker)) +
  geom_boxplot(alpha = 0.5) +
  labs(x = "Configuration", y = "Mean percentage correct") +
  scale_y_continuous(label = function(x) {return(paste(x*100, "%"))}) +
  scale_x_discrete(limits=rev) +
  theme_ggstatsplot() +
  theme(legend.position = "none",
        axis.title = element_text(color = "darkred", size = 14)) +
  scale_fill_brewer(palette="Dark2")

# One-way ANOVA for construction
mp_const <- nerrdata |>
  group_by(id, construction) |>
  summarise(n = n(),

```

```

      x = sum(correct),
      xp = x/n(),
      t = asin(sqrt(x/(n()+1))) + asin(sqrt((x+1)/(n()+1))),
      r = 46.47324337*t - 23,
      .groups = "drop_last")

summary(aov(r ~ construction, data = mp_const))

mp_const |>
  ggplot(aes(x = construction, y = xp, fill = construction)) +
  geom_boxplot(alpha = 0.5) +
  labs(x = "Configuration", y = "Mean percentage correct") +
  scale_y_continuous(label = function(x) {return(paste(x*100, "%"))}) +
  scale_x_discrete(limits=rev) +
  theme_ggstatsplot() +
  theme(legend.position = "none",
        axis.title = element_text(color = "darkred", size = 14)) +
  scale_fill_brewer(palette="Dark2")

#Response Time Analysis
#One-way ANOVA for correctness
summary(aov(Reaction.Time ~ correct, data = nerrdata))

nerrdata |>
  ggplot(aes(x = conf, y = Reaction.Time)) +
  geom_boxplot(mapping = aes(fill = correct), alpha = 0.5) +
  labs(x = "Configuration", y = "Mean response time", fill = "Response") +
  scale_y_log10(label = function(x) {return(paste(x, "ms"))}) +
  scale_x_discrete(limits=rev) +
  theme_ggstatsplot() +
  theme(axis.title = element_text(color = "darkred", size = 14)) +
  scale_fill_brewer(palette="Dark2", labels = c("Incorrect", "Correct"))

#Removing incorrect results from dataset
corrdata <- filter(nerrdata, correct == TRUE)

#One-way ANOVA for speaker
n.data <- corrdata |>
  filter(speaker %in% c("e","p"))
summary(aov(Reaction.Time ~ speaker, data = n.data))

d.data <- corrdata |>
  filter(speaker %in% c("m","s"))
summary(aov(Reaction.Time ~ speaker, data = d.data))

n.data |>
  ggplot(aes(x = speaker, y = Reaction.Time, fill = speaker)) +
  geom_boxplot(alpha = 0.5) +
  labs(x = "Speaker", y = "Response Time") +
  scale_x_discrete(limits=rev) +
  scale_y_log10(label = function(x) {return(paste(x, "ms"))}) +
  theme_ggstatsplot() +
  theme(legend.position = "none",
        axis.title = element_text(color = "darkred", size = 14)) +
  scale_fill_brewer(palette="Dark2")

d.data |>
  ggplot(aes(x = speaker, y = Reaction.Time, fill = speaker)) +
  geom_boxplot(alpha = 0.5) +

```



```

labs(x = "Speaker", y = "Response Time") +
scale_x_discrete(limits=rev) +
scale_y_log10(label = function(x) {return(paste(x, "ms"))}) +
theme_ggstatsplot() +
theme(legend.position = "none",
      axis.title = element_text(color = "darkred", size = 14)) +
scale_fill_brewer(palette="Dark2")

#Construction effect analysis
Anova(aov(Reaction.Time ~ construction, data = corrrdata), type=2)
corrrdata |>
  ggplot(aes(x = construction, y = Reaction.Time, fill = construction)) +
  geom_boxplot(alpha =0.5) +
  labs(x = "Construction", y = "Response time", fill = "Construction") +
  scale_y_log10(label = function(x) {return(paste(x, "ms"))}) +
  theme_ggstatsplot() +
  theme(axis.title = element_text(color = "darkred", size = 14),
        axis.text.x = element_blank(),
        axis.title.x = element_blank())

p_const <- mp_const |>
  group_by(construction)|>
  summarize(correctness = mean(xp))
mrt_const <- corrrdata |>
  group_by(construction) |>
  summarise(mean.reaction.time = mean(Reaction.Time))
const_data <- inner_join(p_const, mrt_const)

const_data|>
  arrange(correctness)
const_data|>
  arrange(desc(mean.reaction.time))

#Linear mixed-effect regression
library(lme4)
library(merTools)
library(lmerTest)
library(MuMIn)

corr_model <- lmer(Reaction.Time ~
                  phonology + morphology + (1 | id) + (1 | construction / index),
                  data = corrrdata)
summary(corr_model)
fixef(corr_model)
plot(corr_model)
plotRESim(RESim(corr_model))
plotFESim(FESim(corr_model))
r.squaredGLMM(corr_model)

ggcoefstats(corr_model, exclude.intercept = TRUE,
            title = "lmer Linear mixed-effects regression coefficients",
            ggtheme = theme_ggstatsplot())

library(ggpubr)
ggqqplot(corrrdata$Reaction.Time)
shapiro.test(corrrdata$Reaction.Time)

#Robust linear mixed-effect regression

```

```

library(robustlmm)
library(insight)

getR2 <- function(modelin) {
  var.fix <- get_variance_fixed(modelin)
  var.ran <- get_variance_random(modelin)
  var.res <- get_variance_residual(modelin)
  R2m = var.fix/(var.fix+var.ran+var.res)
  R2c = (var.fix+var.ran)/(var.fix+var.ran+var.res)

  c("R2m" = R2m, "R2c" = R2c)
}

rcorr_model <- rlmer(Reaction.Time ~
  phonology + morphology + (1 | id) + (1 | construction / index),
  data = corrrdata)
summary(rcorr_model)
fixef(rcorr_model)
plot(rcorr_model)
getR2(rcorr_model)

#Log transformed data
corr_log_model <- lmer(log(Reaction.Time) ~
  phonology + morphology + (1 | id) + (1 | construction / index),
  data = corrrdata)
summary(corr_log_model)
fixef(corr_log_model)
plot(corr_log_model)
getR2(corr_log_model)

ggqqplot(log(corrrdata$Reaction.Time))

rcorr_log_model <- rlmer(log(Reaction.Time) ~
  phonology + morphology + (1 | id) + (1 | construction / index),
  data = corrrdata)
summary(rcorr_log_model)
fixef(rcorr_log_model)
plot(rcorr_log_model)
getR2(rcorr_log_model)

models <- list(
  "lmer" = corr_model,
  "rlmer" = rcorr_model,
  "lmer(log)" = corr_log_model,
  "rlmer(log)" = rcorr_log_model
)

library(modelsummary)
library(tinytable)
modelsummary(models, stars = TRUE, gof_omit = "IC|Adj|F|RMSE|Log") |>
  group_tt(j = list("Linear" = 2:3, "Log-Linear" = 4:5))

modelplot(models[1:2], coef_omit = "Intercept|SD",
  title = "Coefficient estimates and 95% confidence intervals for linear models")
modelplot(models[3:4], coef_omit = "Intercept|SD",
  title = "Coefficient estimates and 95% confidence intervals for log-linear models")

#Equivalence test
diff_corrrdata <- corrrdata |>

```

```

mutate(pmsum = phonology + morphology,
       pmdif = phonology - morphology)
corr_log_diff_model <- lmer(log(Reaction.Time) ~
                           pmsum + pmdif + (1 | id) + (1 | construction / index),
                           data = diff_corrdata )
summary(corr_log_diff_model)

corr_log_comp_model <- lmer(log(Reaction.Time) ~
                           I(phonology + morphology) + (1 | id) + (1 | construction / index),
                           data = corrrdata )
anova(corr_log_model, corr_log_comp_model)

```